CISC7204 Data Science and Data Visualization

Semester Project · Team Project Proposal

The Power-Hungry World: Comparing Global Electricity Consumption Across Sectors

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1. Project Title and Research Question

Title: The Power-Hungry World: Comparing Global Electricity Consumption Across Sectors

Research question: How has electricity consumption changed across the world's countries and regions between 1990 and 2022, and which countries and sectors account for the largest shares of that consumption?

Three sub-questions make the question measurable, and each is answered from the same harmonised table:

- **Change over time —** how has electricity consumption grown by region and in the largest countries, and which regions grew fastest?
- **Country shares —** which countries account for the largest part of world consumption, and has that changed between 1990 and 2022?
- **Sector shares —** how is consumption split between industry, residential, commercial, transport and other uses, worldwide and by region, and how has that split moved?

The question is answerable from public data with descriptive statistics and a small number of tests, which is what makes it suitable for statistical and data-science inquiry rather than for speculation about causes.

2. Project Purpose

Electricity is the fastest-moving part of the world energy system, yet the public conversation about it usually cites a single global number. This project replaces that number with a decomposition: which regions, which countries and which sectors drove the change between 1990 and 2022, and how much of the movement is measured rather than estimated. The output is a defensible description of where consumption sits today and how the balance shifted over three decades.

Outcomes. The team will exercise the full data-science workflow on three genuinely awkward public sources: acquisition with licence constraints, harmonisation of three units and three sector taxonomies, reconciliation of country spellings across sources, cross-source validation, growth-rate and share arithmetic, and figure production for a non-specialist reader.

Skills. pandas joins and reshaping across mismatched country lists; unit conversion between TJ, PJ and TWh; compound annual growth rates, index series and contribution-to-growth decomposition; cross-source reconciliation and reporting-bias checks; matplotlib and seaborn panels for trends and shares; reproducible notebooks with pinned environments.

Knowledge. energy economics and structural change across three decades; the difference between generation and final consumption; why observational national statistics support description rather than causal claims; how statistical agencies estimation practices shape what a comparison can support.

3. Data Context

Region and domain. The geographic region is global: 218 countries and territories, aggregated to the IEA regions for the regional view. The domain category is energy economics, specifically final electricity consumption by end-use sector. Three sources are used, each covering the other's gaps:

Source	Coverage	Years	Unit	Sector detail	Access and licence	Role
IEA World Energy Balances Highlights, 2026 edition	54 countries and 10 aggregates incl. World	1971–2024 (2025 provisional)	PJ	industry, transport, residential, commercial and public services	Free IEA account; raw file must not be redistributed	Primary: sectors, world and regional totals
UN Statistics Division, Energy Balance dataflow DF_UNData_EnergyBalance	218 countries and territories; no aggregates; 4 combined entries	1990–2023	TJ	industry, transport, households, commerce and public services	Public API and CSV, no account; free to copy with UNSD cited	Primary: country totals and rankings
Our World in Data energy data (Ember and Energy Institute)	213 countries with demand data; 6 territories absent 2000–2022	2000–2022	TWh	none — totals, per-person values and generation mix only	Free CSV, XLSX and JSON	Secondary context and generation mix only

KAPSARC electricity consumption by sectors was reviewed and dropped: it covers Saudi Arabia only, which cannot answer a global question.

Definitions. Consumption means final electricity consumption, the electricity delivered to end users; in the UN data, power-plant own use and grid losses are separate lines outside this figure. The window is 1990 to 2022: the UN series

begins in 1990, and 2023 carries a higher share of estimates. The reporting unit is TWh, where one TWh equals 3.6 PJ, which equals 3 600 TJ. Data centres, cryptocurrency mining and demand forecasting are out of scope, because no source in this set reports them separately.

Data quality and limitations

- **Internal consistency —** the IEA 2022 world total for final electricity consumption is 88,859 PJ, about 24,700 TWh, and its sector flows add up to that total for 2021–2024 within rounding.
- **Cross-source disagreement —** the two main sources disagree for some large countries. UN/IEA total ratios are 0.77 for Indonesia, 1.09 for China, 0.98 for India and 0.95 for Korea, and neither gap is explained. Nine other large countries agree within 3 %, and five of those match exactly on every sector — which suggests both sources draw on the same national reports, so agreement is weak evidence of accuracy.
- **Uneven coverage —** UN transport is reported for 98 of 218 areas in 2022 (81 in 2012, 102 in 2023), so a rising sector trend can partly reflect more countries reporting. 17.3 % of 2022 UN values are flagged as estimates, against 12.9 % in 2012 and 20.9 % in 2023, and the meaning of the flag is assumed rather than confirmed.
- **Definitional edges —** four UN entries combine territories (France with Monaco, Italy with San Marino, Norway with Svalbard and Jan Mayen, Switzerland with Liechtenstein), and China has no commerce row, with its 4,363 PJ "other" row possibly absorbing it.
- **Licence and reproducibility —** the IEA licence forbids redistributing the raw file, so each member downloads their own copy and the repository holds code, instructions and a derived extract only. That constraint is why OWID, downloadable in one request, anchors reproducibility for the totals and generation mix.

4. Project Tasks and Deliverables

Preparation turns three differently shaped files into one comparable table of country, year, sector and TWh. Raw files stay untouched; the UN download is kept as a dated CSV; the IEA file is read from its TimeSeries sheet with ".." treated as missing and the provisional 2025 column dropped. A missing sector row is treated as not separately reported, never as zero. The work then runs in six steps:

#	Task	Method and tools	Output
1	Acquire the three sources and freeze them	IEA Excel (each member downloads their own copy), UN CSV from the SDMX API, OWID CSV; date-stamp each file	Three untouched raw files plus a dated manifest
2	Harmonise into one analysis table	Replace UN area codes with names, convert TJ→PJ→TWh, reconcile country spellings across sources, group sectors into the four agreed groups, carry the estimate flag	Table of country, year, sector, TWh, estimate-flag, source
3	Sub-question 1 — change over time	Totals by region and largest country 1990–2022; absolute change, compound annual growth rate, index with 1990 = 100; decade growth as well as the full window	Trend table plus a small-multiple line-figure set
4	Sub-question 2 — country shares	Each country's share of world consumption in 1990, 2000, 2010 and 2022; top ten and rank changes; top-five and top-ten concentration; world total cross-checked against the IEA World row	Ranking table plus a share figure
5	Sub-question 3 — sector shares	Share of each group in world and regional totals, 1990 vs 2022 in percentage points; each sector's growth rate and contribution to total growth; IEA primary, UN as second view	Sector table plus a stacked-share figure
6	Validation, generation mix and reporting	Run the nine validation checks; add the OWID generation-mix panel; assemble the report, the dashboard and the video	Validated results, four figures, dashboard, report and video

Deliverables. Executed Jupyter notebook reproducing every number; four figures (regional trend, country share shift, sector share change, generation mix); one interactive dashboard; this proposal; and the five-minute introductory video.

Data preparation steps

The steps run in this order, and each is visible in the notebook:

- Keep the raw files untouched. The UN download is saved as a dated CSV; each member downloads their own copy of the IEA file, because the licence forbids redistribution.
- Load the IEA workbook from its TimeSeries sheet with the real header row, treat the text value ".." as missing, drop the helper columns, and drop the provisional 2025 column.
- Load the UN file and replace numeric area codes with names from the official code list, labelling the four combined entries clearly.

- Convert units: UN values from TJ to PJ (divide by 1,000), then both sources from PJ to TWh (divide by 3.6).
- Match country names across sources — China, Korea, Turkey, Czechia, Slovakia, the United States and the United Kingdom are spelled differently, and a raw join would silently drop them.
- Group sectors identically in both sources: industry; transport; residential (households); and services and other (commerce and public services, agriculture, other).
- Treat a missing sector row as not separately reported, never as zero — UN totals equal the sum of reported parts, so a missing row hides no amount but may be counted inside another.
- Build the UN world total by summing its 218 areas, since it has no World row, and compare it with the IEA World row.
- Prepare OWID for context only: keep rows with an ISO code (aggregates have none) and the years 2000 to 2022.
- Carry the estimate flag through the analysis, so any result can be re-run with estimated values excluded.

Analysis by sub-question

Every result is reported with its source, its year range, its unit and the share of values that are estimates.

- **Change over time —** total consumption in TWh by region and for the largest countries for every year from 1990 to 2022; absolute change, compound annual growth rate, and an index with 1990 equal to 100; growth by decade as well as over the full window, because the 2020 pandemic dip bends any single rate. Regions come from the IEA aggregates; country detail from the UN data.
- **Country shares —** each country's share of the world total in 1990, 2000, 2010 and 2022, the top ten and how their ranks changed, and the combined share of the top five and top ten to show concentration; the world total is the sum of the UN areas, cross-checked against the IEA World row.
- **Sector shares —** each grouped sector's share of world and regional totals, 1990 against 2022, with the change in percentage points; each sector's growth rate and its contribution to total growth. The IEA file is primary for sectors and the UN data gives a second view; where they differ, both are reported.

Additional analysis

A. Generation mix — Two questions sit beside the main one and do not change it. The first uses OWID's electricity columns, which cover 2000 to 2022: the fuel-by-fuel generation mix — coal, gas, oil, nuclear, hydro, solar, wind, biofuel and other renewables — compared for the world, the regions and the largest countries in 2000 and 2022, plus each fuel's growth rate. Generation is not consumption: it includes losses and trade, so it stays out of the sector shares and is shown side by side with them, labelled as such.

B. Why consumption changed — the data is observational and carries no prices, policies, technology costs or weather, so it can describe patterns but cannot show that one thing caused another. The team relates consumption growth to population growth and to GDP per person for the largest countries, and reports the result as consistent with an explanation, never as proof; the data a real test would need is stated explicitly.

Resolved scope decisions

The nine decisions the draft left open are settled as follows, so every downstream number has a fixed definition.

#	Decision	Value
1	Research question	The wording below and its three sub-questions, as drafted
2	Window	1990–2022
3	Sector grouping	Four groups: industry; transport; residential; services and other
4	Unit	TWh (1 TWh = 3.6 PJ = 3 600 TJ)
5	Regions	IEA aggregates for world and regional totals; UN for country detail; no self-built mapping
6	Extensions	Include the generation-mix panel (OWID, 2000–2022); include only the simple comparison of consumption growth against population and GDP growth, not the three-way decomposition
7	Roles	As recorded in the 1st meeting minutes (Section 6)
8	Brief compliance	Checked line by line against the brief by UN CHI KIT
9	Environment	python 3.14.7, pandas 3.0.3, numpy 2.4.6, openpyxl 3.1.5, requests 2.34.2, notebook 7.5.7, ipykernel 6.31.0, matplotlib 3.11.0, seaborn 0.13.2, pinned in requirements.txt

Environment. The shared environment is pinned so every member reproduces the same figures: python 3.14.7, pandas 3.0.3, numpy 2.4.6, openpyxl 3.1.5, requests 2.34.2, notebook 7.5.7, ipykernel 6.31.0, matplotlib 3.11.0, seaborn 0.13.2. Each notebook prints its versions in the first cell, and the pins live in requirements.txt beside the code.

5. Success Criteria

Nine checks decide whether the final work is good enough. Each is observable, and each is run before submission rather than asserted afterwards.

Check	What makes it pass
Data quality	Missing values, estimates and coverage gaps stated for every table
Reproducibility	The notebook runs clean from a restart on a fresh environment
Calculation correctness	Sector shares sum to 100 %; TJ, PJ and TWh conversions consistent
Figure clarity	Every figure carries title, units, source and years
Citations and licences	Every source and licence condition covered, IEA cited in its required form
Brief compliance	Checked line by line against the assignment brief
Reporting bias	Sector trends repeated using only countries that report in every year
Estimate sensitivity	Key results re-run with estimated values excluded
Traceability	Every sentence in the report ties to a number in the outputs

6. Team Collaboration

Roles are as recorded in the 1st meeting minutes of 28 October 2026. The four roles the brief names explicitly are held by the members below, and the remaining work is divided so that every task has one accountable owner.

Member	Student ID	Role(s)
UN CHI KIT	mc653081	Team coordinator and project tracker
KELDEN JAMTSHO	mc664183	Project analyst and quality reviewer
GAN RUIYAN	mc664801	Lead project developer
HE YIFEI	mc664916	Team reporter and team liaison
SHENG TIAN	mc664961	Second project developer
TANG JING	mc666256	Second project analyst and team scribe

Designated roles. the team is coordinated by UN CHI KIT, who also keeps the progress tracker; the minutes and the shared repository are kept by TANG JING as team scribe; and HE YIFEI is the liaison with the instructor, relaying requirements and feedback. Major decisions are taken by the team together and recorded in the minutes.

Working rules.

- one shared repository, files filed under the agreed naming convention, and every change to code, filters or design documented in Git.
- every number in a deliverable comes from the notebook; nothing is typed by hand.
- the quality reviewer signs off each deliverable before it is submitted, and the coordinator ticks off each member's individual submission by name.

7. Timeline

The schedule begins on the proposal submission date and completes all project activity within four weeks, ending on 3 December 2026; the collaboration record follows on 5 December. One online meeting is held each week.

Week	Dates	Activity	Owner
1	5–11 Nov 2026	Acquire and freeze the three sources; harmonise country names and sector groups; build the UN world total and reconcile it with the IEA World row	Analyst (Kelden) + developer (Gan)
2	12–18 Nov 2026	Sub-questions 1 and 2: growth rates, index series, regional totals, country shares and concentration	Analyst + second analyst (Tang)
3	19–25 Nov 2026	Sub-question 3: sector shares and contributions to growth; generation-mix panel; first drafts of the four figures	Developer (Sheng) + analyst
4	26 Nov–3 Dec 2026	Report, dashboard, rehearsal, and the collaboration record (5 Dec); final validation pass before submission	Reporter (He Yifei) + coordinator (Un Chi Kit)

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